

INTERNSHIP REPORT
ON
WEB DEVELOPMENT INTERNSHIP

Project: SMS Spam Detection Using Machine Learning

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.



By

LEKISHA

Roll Number: 25SCS1003001466

Batch: 2025–2029

CANDIDATE'S DECLARATION

I, LEKISHA, do hereby declare that the internship report titled “Web Development Internship at Codec Technologies Pvt. Ltd.”, including the project “SMS Spam Detection Using Machine Learning”, has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

The internship was undertaken in the role of Web Developer Intern at Codec Technologies Pvt. Ltd. from 01 July 2026 to 30 July 2026. The work presented in this report reflects my understanding, practical learning, technical activities, and project experience gained during the internship.

I further declare that this report has been prepared for academic purposes and has not been submitted elsewhere for the award of any other degree, diploma, or qualification. All references used in the preparation of this report have been appropriately acknowledged.

Signature: _____

Student Name: LEKISHA

Roll No.: 25SCS1003001466

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Codec Technologies Pvt. Ltd. for providing me with the opportunity to undertake a one-month internship as a Web Developer Intern.

This internship provided me with valuable exposure to a professional working environment and helped me understand how theoretical concepts are applied to practical projects. During the internship, I gained experience in programming, data handling, problem-solving, project development, debugging, documentation, and understanding the overall software development workflow.

I am especially thankful to the project head and mentors for their guidance, support, and feedback throughout the internship. Their suggestions helped me improve my technical understanding and approach problems in a more structured manner.

I am also grateful to my faculty members and IILM University for encouraging practical learning and providing the academic support required to complete this internship successfully.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of my internship and this report.

Signature: _____

Student Name: LEKISHA

Roll No.: 25SCS1003001466

INTERNSHIP COMPLETION CERTIFICATE

The internship certificate issued by Codec Technologies Pvt. Ltd. is attached below as documentary proof of successful completion of the internship.



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4. PROJECT DESCRIPTION

4.1 Introduction

An internship provides students with an opportunity to gain practical exposure beyond classroom learning. It helps bridge the gap between theoretical knowledge and real-world implementation by allowing students to work on practical tasks and understand professional development processes.

I completed a one-month internship at Codec Technologies Pvt. Ltd. from 01 July 2026 to 30 July 2026 as a Web Developer Intern. The internship was conducted in a Pan India, Hybrid mode and was approved under AICTE and ICAC.

During the internship, I was introduced to practical software development practices and worked on a project titled “SMS Spam Detection Using Machine Learning.” Although the project involved Machine Learning and Natural Language Processing concepts, the internship provided broader exposure to development practices, Python programming, data processing, debugging, documentation, and systematic project implementation.

The internship also helped me understand how software projects are developed and managed in a professional environment. It provided an opportunity to apply programming concepts to practical problems, understand project requirements, work with real-world datasets, develop structured solutions, debug and improve implementations, compare technical approaches, understand testing and evaluation, improve technical documentation, and develop professional communication and problem-solving skills.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. provided the internship environment in which I gained practical exposure to software and technology-related activities. The organization provided an internship environment focused on practical learning and project-based development.

According to the internship offer letter, Codec Technologies is described as a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. It delivers dedicated IT and business consultancy services across 27+ countries, with a focus on global innovation and strategic growth.

My designated role during the internship was Web Developer Intern. As a Web Developer Intern, I was exposed to software development practices, programming, problem-solving, project implementation, and documentation. The assigned project also introduced me to Machine Learning-based development through the implementation of an SMS spam classification system.

The internship was completed from 01 July 2026 to 30 July 2026 in Pan India, Hybrid mode, with reporting to assigned project head(s).

4.3 Problem Statement

SMS messages are commonly used for communication, but unwanted promotional and fraudulent messages can create security and privacy concerns. Manually identifying such messages is time-consuming and may lead to incorrect decisions.

The problem addressed by this project was to develop an automated system capable of classifying SMS messages into two categories: Ham, representing legitimate messages, and Spam, representing unwanted or suspicious messages.

Since Machine Learning models cannot directly process raw text, the project required a complete text-classification pipeline involving data cleaning, text preprocessing, feature extraction, model development, comparison, ensemble learning, and evaluation.

4.4 Project Objectives

- Develop an effective SMS spam classification system capable of distinguishing legitimate and spam messages.
- Strengthen practical programming skills and understanding of Python.
- Work with a real-world SMS dataset and understand its characteristics.
- Perform data cleaning, preprocessing, and exploratory data analysis.
- Apply Natural Language Processing techniques such as tokenization, stopwords removal, and stemming.
- Use TF-IDF to convert text messages into numerical feature representations.
- Implement and compare multiple Machine Learning classification algorithms.
- Evaluate models using accuracy, precision, recall, and F1-score.
- Apply ensemble learning and identify an effective final model.
- Develop systematic problem-solving, debugging, documentation, and professional communication skills.

4.5 Scope of the Project

The scope of the project covers the development of an SMS text-classification pipeline that can distinguish between Ham and Spam messages. The work includes dataset preparation, cleaning, text preprocessing, exploratory analysis, TF-IDF feature extraction, Machine Learning model training, model comparison, ensemble learning, and final evaluation.

The system can potentially be applied to mobile messaging applications, SMS filtering systems, communication platforms, fraud detection systems, promotional message filtering, and automated message classification.

The project also provides a foundation for future integration into a web or mobile application where a user can enter an SMS message and receive an immediate prediction. A web-based interface would connect the Machine Learning component with the Web Development focus of the internship.

The current scope has limitations. Model performance depends on the quality and characteristics of the training dataset. Spam language and patterns change over time, effectiveness may vary for languages or writing styles not represented in the training data, and large-scale real-time deployment would require additional engineering.

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Python	Programming and implementation
Pandas	Data manipulation
NumPy	Numerical operations
Scikit-learn	Machine Learning, feature extraction, training and evaluation
NLTK	Natural Language Processing operations
Matplotlib	Data visualization
Seaborn	Exploratory data analysis
TF-IDF	Text feature extraction
Jupyter Notebook	Development and experimentation
Git/GitHub	Version control and documentation

Python was used as the primary programming language because of its simple syntax and extensive libraries. Pandas was used for dataset loading, cleaning, filtering, analysis, and manipulation. NumPy supported numerical operations and efficient data processing. Scikit-learn was used for dataset splitting, feature extraction, model training, model evaluation, classification algorithms, and ensemble learning.

NLTK was used for Natural Language Processing operations such as tokenization, stopword removal, and stemming. Matplotlib and Seaborn were used for visualization and exploratory data analysis. Jupyter Notebook supported development and experimentation, while Git/GitHub supported version control and documentation.

4.7 System Architecture

The overall system follows a sequential text-classification workflow:

- SMS Dataset → Data Understanding → Data Cleaning
- Text Preprocessing → Tokenization, Stopword Removal, Stemming
- TF-IDF Feature Extraction → Train/Test Split
- Machine Learning Models → SVC, Multinomial Naive Bayes, Extra Trees
- Soft Voting Classifier → Evaluation → Spam/Ham Classification

The dataset initially contained 5,572 records. After cleaning and removing duplicate records, the final dataset contained 5,169 records, comprising 4,516 Ham messages and 653 Spam messages.

The cleaned text is transformed into numerical vectors using TF-IDF. These feature vectors are then supplied to different Machine Learning algorithms. The selected classifiers contribute to a Soft Voting Classifier, whose predictions are evaluated using standard classification metrics.

This architecture separates data preparation, feature extraction, model development, ensemble learning, and evaluation into systematic stages, making the project easier to understand, test, debug, and improve.

4.8 Methodology

4.8.1 Dataset and Data Understanding

The SMS dataset initially contained 5,572 records. After cleaning and removing duplicate records, the final dataset contained 5,169 records. The cleaned dataset consisted of 4,516 Ham messages and 653 Spam messages.

The dataset contained significantly more legitimate messages than spam messages. This class imbalance is important because a classification model should not simply achieve high accuracy by predicting the majority class. Therefore, multiple evaluation metrics were considered.

4.8.2 Data Preprocessing

Data preprocessing was one of the most important stages of the project. Raw SMS messages contain punctuation, unnecessary characters, different capitalization, stopwords, and other information that may not be useful for classification.

The messages were cleaned to remove unnecessary elements and standardize the text. Tokenization was then used to divide sentences into individual words or tokens. Common stopwords that provide limited classification information were removed. Stemming was used to reduce words to their basic or root forms, such as converting “playing” and “played” toward “play”.

4.8.3 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the characteristics of the dataset. The analysis showed 4,516 Ham messages and 653 Spam messages, demonstrating the imbalance between the two classes.

Spam messages commonly contain promotional language, offers, urgency, prizes, financial terms, or calls to action. Legitimate messages generally represent normal personal, informational, or service-related communication. EDA helped in understanding the dataset, identifying class imbalance, understanding text characteristics, identifying useful patterns, and preparing the data for Machine Learning.

4.8.4 Feature Extraction Using TF-IDF

Machine Learning models cannot directly process raw text. Therefore, the text messages were converted into numerical features using TF-IDF, which stands for Term Frequency–Inverse Document Frequency.

The basic idea is: $\text{TF-IDF} = \text{Term Frequency} \times \text{Inverse Document Frequency}$. Words that are common throughout the dataset receive lower importance, while distinctive words receive higher importance.

TF-IDF was suitable because it converts text into numerical features, captures important words, reduces the impact of extremely common terms, works effectively with traditional Machine Learning algorithms, and is computationally practical for text classification.

4.8.5 Machine Learning Model Development

Multiple Machine Learning algorithms were implemented and evaluated. The models included Naive Bayes, Extra Trees, Random Forest, Support Vector Classifier, Logistic Regression, AdaBoost, XGBoost, and ensemble models.

Naive Bayes was used as a probabilistic classifier suitable for text classification and TF-IDF features. Random Forest combines multiple decision trees, while Extra Trees introduces additional randomness while creating decision trees. Support Vector Classification attempts to identify an effective decision boundary and is suitable for high-dimensional TF-IDF feature spaces. Logistic Regression provided a strong baseline for binary classification. AdaBoost combined weak learners into a stronger classifier, while XGBoost was included as an efficient gradient boosting approach.

4.8.6 Model Comparison and Evaluation

The models were trained and evaluated to determine which approach provided the most effective classification performance. The major evaluation metrics considered were accuracy, precision, recall, and F1-score.

Accuracy measures the proportion of correctly classified messages. Precision measures how many messages predicted as spam were actually spam. Recall measures how many actual spam messages were successfully detected. F1-score combines precision and recall into a single metric and is useful when class distributions are not perfectly balanced.

4.8.7 Ensemble Learning and Final Model

After evaluating individual Machine Learning algorithms, ensemble learning techniques were considered. The best recorded result was achieved using a Soft Voting Classifier combining Support Vector Classifier (SVC), Multinomial Naive Bayes, and Extra Trees.

The ensemble approach combines predictions from multiple models rather than relying on only one classifier. The best recorded performance was 98.16% accuracy and 99.17% precision. The high precision indicates that the model was effective at correctly identifying messages predicted as spam.

4.8.8 Development Process

- Step 1 – Dataset Preparation: The dataset was loaded and inspected.
- Step 2 – Data Cleaning: Duplicate and unnecessary records were removed.
- Step 3 – Text Processing: SMS messages were cleaned, tokenized, and processed.
- Step 4 – Feature Extraction: TF-IDF was applied to convert text into numerical vectors.
- Step 5 – Model Training: Multiple Machine Learning algorithms were trained.
- Step 6 – Model Evaluation: The models were compared using classification metrics.
- Step 7 – Ensemble Learning: A Soft Voting Classifier was created using selected models.
- Step 8 – Final Evaluation: The final model achieved the best recorded performance of 98.16% accuracy and 99.17% precision.

4.9 Expected Outcomes

The expected outcomes of the internship and project were to gain practical exposure to software development and project implementation, strengthen Python programming skills, understand data handling and preprocessing, learn Natural Language Processing and TF-IDF feature extraction, implement and evaluate Machine Learning classification algorithms, and develop a systematic approach to debugging and documentation.

The project achieved a strong recorded classification performance, with the Soft Voting Classifier obtaining 98.16% accuracy and 99.17% precision. The internship also improved professional skills including communication, time management, independent learning, problem-solving, and technical documentation.

The project can be further extended through a larger and more diverse dataset, real-time application integration, advanced NLP techniques such as word embeddings, deep learning and transformer-based models, multilingual spam detection, and a user-friendly web-based interface.

4.10 Certificates of Completion and Other Communication Mail Proof

The following documentary proofs are attached: (1) Internship Completion Certificate issued by Codec Technologies Pvt. Ltd.; and (2) Internship Offer Letter containing the designation, duration, mode, and reporting details.

The offer letter confirms the designation as Web Developer Intern, Pan India Hybrid mode, internship duration from 01/07/2026 to 30/07/2026, and reporting to assigned project head(s).

Internship Offer Letter

INTERNSHIP OFFER LETTER

Dear Lekisha,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Web Developer Intern
- Location & Mode: Pan India, Hybrid
- Duration : 01/07/2026 to 30/07/2026
- Reporting to : Assigned Project Head(s)

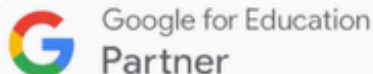
This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 01/07/2026


Dr. Anurag Shrivastava
Talent Acquisition Manager

5. BIBLIOGRAPHY / REFERENCES

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2. Pandas Documentation.
3. NumPy Documentation.
4. Scikit-learn Documentation.
5. NLTK Documentation.
6. Matplotlib Documentation.
7. Seaborn Documentation.
8. Machine Learning and Natural Language Processing learning resources.
9. Internship training material provided during the internship.
10. SMS Spam Classification dataset and project resources.